llnl_c1/lulesh · lulesh.cc · llm_pragmas_v5 · claude-haiku-4-5-20251001 · BROKEN · RIGHT = attempt 2 of 71, rejected by the gate at 'correctness'

−1 / +38 lines of 1967 · 1 parallel region(s) added · 32 of 71 candidate(s) kept · RESTRUCTURING (28 code line(s) added, 1 removed) · restructuring level 0

BEFORE

```
1960   #ifdef SEDOV_SYNC_POS_VEL_LATE
1961   #endif
1962
1963      LagrangeElements(domain, domain.numElem());
1964
1965      CalcTimeConstraintsForElems(domain);
1966
1967   }
```

ATTEMPT 2 OF 71, REJECTED BY THE GATE AT 'CORRECTNESS'

file

```
1960   #ifdef SEDOV_SYNC_POS_VEL_LATE
1961   #endif
1962
1963      LagrangeElements(domain, domain.numElem());
1964
1965      /* Parallelize time constraint calculation with explicit reductions */
1966      domain.dtcourant() = 1.0e+20;
1967      domain.dthydro() = 1.0e+20;
1968
1969      Real_t qqc2 = Real_t(64.0) * domain.qqc() * domain.qqc();
1970      Real_t dtcourant_min = 1.0e+20;
1971      Real_t dthydro_min = 1.0e+20;
1972
1973      /* Flatten to iterate over all elements for better work distribution and re
1974   #pragma omp parallel for reduction(min:dtcourant_min, dthydro_min) schedule(dy
1975      for (Index_t elemIdx = 0; elemIdx < domain.numElem(); ++elemIdx) {
1976         Index_t indx = elemIdx;
1977
1978         /* Courant constraint calculation */
1979         Real_t dtf = domain.ss(indx) * domain.ss(indx);
1980         if (domain.vdov(indx) < Real_t(0.)) {
1981            dtf = dtf + qqc2 * domain.arealg(indx) * domain.arealg(indx)
1982                     * domain.vdov(indx) * domain.vdov(indx);
1983         }
1984         dtf = SQRT(dtf);
1985         dtf = domain.arealg(indx) / dtf;
1986         if (domain.vdov(indx) != Real_t(0.)) {
1987            if (dtf < dtcourant_min) {
1988               dtcourant_min = dtf;
1989            }
1990         }
1991
1992         /* Hydro constraint calculation */
1993         if (domain.vdov(indx) != Real_t(0.)) {
1994            Real_t dtdvov = domain.dvovmax() / (FABS(domain.vdov(indx)) + Real_t(
1995            if (dthydro_min > dtdvov) {
1996               dthydro_min = dtdvov;
1997            }
1998         }
1999      }
2000
2001      domain.dtcourant() = dtcourant_min;
2002      domain.dthydro() = dthydro_min;
2003
2004   }
```

red = removed green = added yellow = parallel region (#pragma omp) right strip = the whole file: green where changed, yellow where a parallel region now is